

Spatial hearing in VR: individualised HRTFs and visual references

RESEARCH INTERN • CERCo, CNRS, TOULOUSE • 2019 • C++, UNITY, HTC VIVE, R

Linked: [first-year master's thesis](#)

Does binaural rendering need to be individualised, and does the visual scene around the listener change how well they localise?

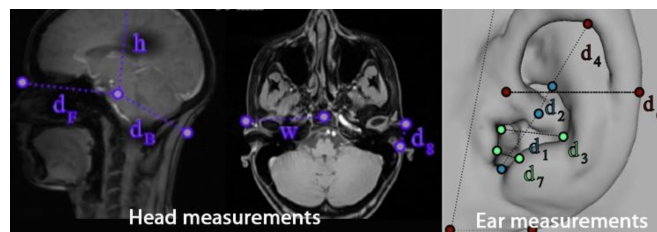
What I built

My first research engineering project: a VR system for spatial hearing, combining binaural rendering over earphones with visual environments built in Unity and presented through an HTC Vive, with head-tracked pointing responses recorded and converted to polar coordinates for analysis.

Experiment 1 - does individualisation matter?

I compared four HRTF sets by measuring the minimum audible angle (2AFC, 1-up/1-down) at three azimuths in 14 normal-hearing listeners: two manikin sets (KEMAR, KU100) and two individualised sets selected from a 43-subject database by matching each listener's anthropometry - head and ear measurements, or ear measurements only. Monaural attenuation conditions were added to simulate unilateral hearing loss.

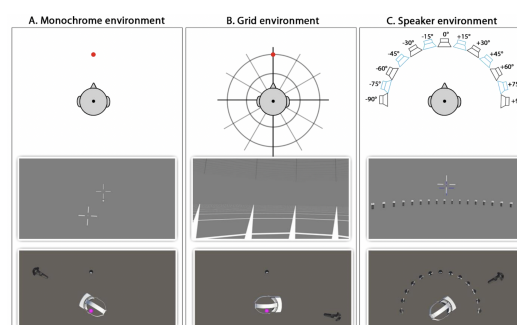
Result: individualisation did not improve spatial resolution. Only KEMAR was significantly worse than the rest; the individualised sets were indistinguishable from KU100. The monaural thresholds matched published values for listeners with unilateral hearing loss, suggesting the paradigm could serve as a low-cost clinical measure of spatial resolution.



Head and ear anthropometric parameters used to match each listener to a database HRTF.

Experiment 2 - does the visual scene matter?

Listeners localised sounds at seven azimuths in three environments of increasing visual structure - monochrome, grid, and visible loudspeakers - with lateral bias and lateral precision error as outcomes. Neither measure varied with the environment. A plausible explanation is that all three scenes shared a fixed 0° reference, which may have been sufficient on its own.



The three environments: monochrome (A), grid (B) and speaker (C), each shown as the theoretical layout, the participant's view and the experimenter's view.